

Nexus-9 Quantum Processing Unit (QPU)

Technical Datasheet v4.2

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Manufacturer: Archaon Systems

1. Architecture Overview

The Nexus-9 QPU introduces the new "Hyper-State" architecture, designed to stabilize qubit coherence at room temperatures for up to 50 milliseconds. Unlike the previous Nexus-8 model, which utilized a purely superconducting loop, the Nexus-9 integrates a hybrid photonic-trap interface. This allows for a 40% reduction in error rates during gate operations.

The system is comprised of three core modules:

- The Core Lattice:** Houses the 128 physical qubits.
- The Control Plane:** Manages microwave pulse generation.
- The Readout Layer:** Converts quantum states into digital signals via the PCIe 6.0 interface.

2. Operating Conditions & Performance

The Nexus-9 requires precise environmental controls to maintain the "Hyper-State." Deviation from these parameters will result in immediate decoherence and potential hardware damage.

Metric	Nominal Range	Maximum Limit (Burst)	Critical Threshold
Core Temperature	15 mK - 20 mK	25 mK (for < 2s)	> 30 mK
Power Consumption	4.5 kW	6.2 kW	> 7.0 kW
Coolant Flow Rate	12.0 L/min	15.0 L/min	< 10.0 L/min
Vibration Tolerance	< 0.05 g	< 0.1 g	> 0.2 g

WARNING: If the **Core Temperature** exceeds the Critical Threshold (> 30 mK), the automated failsafe will trigger the **Liquid Helium Dump** protocol. This process is irreversible and will require a full core replacement.

3. Cooling System Requirements (The "Cryo-Stack")

The cooling system operates in four distinct stages. It is crucial that the stages are initialized in this specific order to prevent thermal shock to the ceramic interposers.

- **Stage 1: Pre-Cooling (Ambient to 4K)**
 - Uses a pulse-tube cryocooler.
 - Duration: 12 hours minimum.
 - *Note: Do not activate the Control Plane during this stage.*
- **Stage 2: Condensation (4K to 1.2K)**
 - Uses the Joule-Thomson expansion valve.
 - Pressure must be maintained at 14 bar.
- **Stage 3: Dilution (1.2K to 100mK)**
 - Circulates the Helium-3/Helium-4 isotope mix.
 - The "Mix Chamber" valve must be opened to 45%.
- **Stage 4: Hyper-Cooling (100mK to 15mK)**
 - Activates the demagnetization reducers.
 - **CRITICAL:** Once Stage 4 begins, the system cannot be powered down for 4 hours.

4. Troubleshooting & Error Codes

The following codes may appear on the Readout Layer during initialization.

- **ERR-404 (Coherence Loss):** Usually caused by vibration exceeding 0.05 g. Check isolation mounts.
- **ERR-502 (Thermal Runaway):** Indicates a failure in Stage 3 Dilution. Check the He-3 isotope levels.
- **ERR-909 (Voltage Spike):** Detected in the Control Plane. Immediate shutdown required.